A

GB1 Deep Mutational Scanning Library

B

DMS Library Code

```python
import poolparty as pp

pp.init()
orf_pool = pp.from_seq(GB1_ORF)
pos = slice(1, 56)

single_pool = orf_pool.mutagenize_orf(        # Single mutant
    num_mutations=1,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='single',
    mode='sequential')

double_pool = orf_pool.mutagenize_orf(        # Pairwise mutant
    num_mutations=2,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='double',
    mode='sequential')

random_pool = orf_pool.mutagenize_orf(        # Random mutant
    mutation_rate=0.1,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='ramdom',
    mode='random',
    num_states=10000)

wt_pool = orf_pool.repeat(times=100)          # WT control

# Combine and generate
dms_pool = pp.stack([single_pool, double_pool, random_pool, wt_pool])
dms_df = dms_pool.generate_library(num_cycles=1, seed=42)
```

C

Output

```
>>> dms_df[design_card_columns]
```

name	position	wt_codon	mut_codon	wt_aa	mut_aa	repeat_idx	seq
single	1	CAG	TTC	Q	F	*None*	ATGTTCTAC...
single	1	CAG	CTG	Q	L	*None*	ATGCTGTAC...
[988 rows]							
pairwise	1, 2	CAG,TAC	TTC,TTC	Q,Y	F,F	*None*	ATGTTCTTC...
pairwise	1, 2	CAG,TAC	TTC,CTG	Q,Y	F,L	*None*	ATGTTCCTG...
[478,686 rows]							
random	3	AAG	ATC	K	I	*None*	ATGCAGTAC...
random	1, 4, 6	CAG,CTG,CTG	TTC,ATG,CAG	Q,L,L	F,M,Q	*None*	ATGTTCTAC...
[10,000 rows]							
wt	*None*	*None*	*None*	*None*	*None*	0	ATGCAGTAC...
wt	*None*	*None*	*None*	*None*	*None*	1	ATGCAGTAC...
[100 rows]							

D

DAG View

```
>>> dms_pool.print_dag()

dms_pool (pool, n=489,774)
└── op[5]:stack [mode=fixed, n=1]
    ├── single_pool (pool, n=988)
    │   └── op[1]:mutagenize_orf [mode=sequential, n=988]
    │       └── orf_pool (pool, n=1)
    │           └── op[0]:from_seq [mode=fixed, n=1]
    ├── double_pool (pool, n=478,686)
    │   └── op[2]:mutagenize_orf [mode=sequential, n=478,686]
    │       └── orf_pool (pool, n=1)
    │           └── op[0]:from_seq [mode=fixed, n=1]
    ├── random_pool (pool, n=10,000)
    │   └── op[3]:mutagenize_orf [mode=random, n=10,000]
    │       └── orf_pool (pool, n=1)
    │           └── op[0]:from_seq [mode=fixed, n=1]
    └── wt_pool (pool, n=100)
        └── op[4]:repeat [mode=sequential, n=100]
            └── orf_pool (pool, n=1)
                └── op[0]:from_seq [mode=fixed, n=1]
```